

1. Grammar G_1

Formula ::= Formula '<->' Formula | ImpTerm
ImpTerm ::= ImpTerm '->' ImpTerm | OrTerm
OrTerm ::= OrTerm '\|' OrTerm | AndTerm
AndTerm ::= AndTerm '\^' AndTerm | Negation
NotTerm ::= '!' Negation | Factor
Factor ::= '(' Formula ')' | 'T' | 'F' | Ident

2. Grammar G_2

Formula ::= ImpTerm '<->' Formula | ImpTerm
ImpTerm ::= OrTerm '->' ImpTerm | OrTerm
OrTerm ::= AndTerm '\|' OrTerm | AndTerm
AndTerm ::= Negation '\^' AndTerm | Negation
Negation ::= '!' Negation | Factor
Factor ::= '(' Formula ')' | 'T' | 'F' | Ident

Design choices, features, issues:

- A lot of the code used was copied from lecture examples (everything following q3)
- Used applicative style for readability
- The grammar G_2 guided writing of code for q6
- parseFormula style was copied directly from lecture examples
- main function to handle IO was copied directly from the first programming assignment